JASIR ABDUL HAMEED

Applied AI Engineer - LLM Applications, Retrieval (RAG) and ML Model Serving

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SUMMARY

Computer Science graduate (B.Tech, 2026) who builds and evaluates LLM applications and ML inference services in Python. Built a public-beta RAG assistant with a measured retrieval benchmark, a FastAPI model-serving API with artifact verification and a written model card, and a QLoRA fine-tuned language model published on Hugging Face.

EDUCATION

B.Tech, Computer Science and Engineering - MES College of Engineering, Kuttippuram, Kerala **2022 - 2026**

TECHNICAL SKILLS

LLMs and retrieval: RAG, Qdrant, BM25, reciprocal rank fusion, BGE embeddings, retrieval evaluation (Hit@k, MRR, nDCG), Gemini/Groq/OpenRouter APIs

Model development: QLoRA, PyTorch, Hugging Face (Transformers, PEFT, TRL), scikit-learn, pandas, NumPy, cross-validation

Backend and MLOps: Python, FastAPI, Pydantic, PostgreSQL, Alembic, pytest, Playwright, Docker, GitHub Actions

Frontend: TypeScript, JavaScript, Next.js, React, HTML, CSS

PROJECTS

Jruva - AI Assistant with Hybrid RAG and Multi-Provider LLM Routing **2026**

Public beta: jruva.in | Benchmark: github.com/jasirjru/jruva-retrieval-eval

- Built and deployed a public-beta AI assistant with a FastAPI backend and a Next.js/TypeScript frontend, supporting streamed chat responses and question answering over uploaded documents.
- Implemented hybrid retrieval that fuses BGE-small dense embeddings in Qdrant with Okapi BM25 through weighted reciprocal rank fusion, with retrieval scoped to each user's own documents.
- Built and published a reproducible retrieval benchmark (Hit@k, MRR, nDCG) on 100 queries (90 answerable, 50 passages): dense retrieval reached MRR 0.91 and Hit@3 98.9% versus 0.81 and 84.4% for BM25; hybrid fusion did not beat dense overall (MRR 0.90) and halved Hit@1 on ambiguous queries.
- Traced a reported 0.36 ms retrieval latency to a pre-warmed embedding cache and a disabled reranker, then re-measured cold dense retrieval at 3.7 ms median (query embedding plus in-memory search, CPU).
- Implemented LLM routing across Gemini, Groq and OpenRouter with retries, circuit-breaker failover, response caching and per-user quotas; CI runs linting, migration checks, backend tests and a TypeScript build.

CustomerIQ - Churn Scoring Inference API **2026**

Live demo: customeriq-qd2v.onrender.com | Source: github.com/jasirjru/CustomerIQ

- Built a FastAPI service that serves a scikit-learn preprocessing pipeline and Random Forest churn scores through single-record and batch endpoints (up to 100 records per request).
- Enforced Pydantic validation of categorical values, numeric bounds, unknown fields and cross-field constraints; the service refuses to start if SHA-256 artifact hashes, model parameters or feature order differ from a versioned manifest.
- Compared five classifiers in development (ROC-AUC 0.79-0.84); wrote a model card documenting uncalibrated scores, an exposed historical holdout and the need for fresh labelled data before any accuracy claim.
- Configured GitHub Actions with pytest, Playwright desktop and mobile browser tests, a Docker image build, a container vulnerability scan and an authenticated inference smoke test.

DomainTune - QLoRA Fine-Tuning for Support-Ticket Triage **2026**

Model: huggingface.co/jasirjru/DomainTune-Qwen2.5-1.5B-Triage | Code and results: github.com/jasirjru/DomainTune

- Fine-tuned Qwen2.5-1.5B with QLoRA (4-bit NF4, LoRA rank 16) using PEFT and TRL to turn raw issue text into a five-field JSON triage record, with a data pipeline for label mapping, LSH near-duplicate removal and 80/10/10 splits.
- Evaluated base vs fine-tuned on a 40-example gold set: valid schema-conformant JSON rose from 22.5% to 92.5%, but no output got all five fields right; error analysis traced a missing stop token, majority-class collapse and label noise as the next fixes.

ACHIEVEMENTS

NASA Space Apps Challenge 2024 - Global Nominee and People's Choice **2024**

- Website developer in six-member Team DarkNova (Thrissur, India local event) for "Galactic Games: Fun in a Microgravity Environment"; built the HTML/CSS/JavaScript site with Three.js 3D models: github.com/jasirjru/dark_nova